

What State-Space Injection Buys in Open-Ended Generation, and What It Costs

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Abstract

A research program on machine novelty has intervened at three depths of a language model (the token distribution, the context, and the weights) and measured the limit of each. This paper opens the fourth, the residual stream during live generation, and asks what an intervention there buys against the written word. We inject model-derived concept directions as guarded, pulsed translations into Qwen3-8B-Base and measure the result under a judge-blind protocol: a blind judge scores surprise, connection and coherence on windows of generated-only text, paired over premises, every confirmatory analysis pre-registered. (i) Against text interruption, the program’s flagship prompt-side intervention, the guarded pulse trails in judged surprise at every matched offset (thirty premises, three offsets judged in both arms: -0.73 [$-1.19, -0.27$]; at the end of the intervention itself the written subject scores 1.2 points higher) and trails in connection and coherence; the pre-registered cell average, which contains the pulse’s best offset and no text counterpart, had shown equivalence within ± 0.75 , with its coherence guard failing. (ii) A meaningful direction carries the effect: sham pulses in random or shuffled directions displace the state more and produce less, and a blind manipulation check finds the injected subject in the text, about half as often as when written. (iii) Pulsing for 32 tokens every 300 beats pressing at equal per-token dose, where the pressed arm degenerates; at equal integrated dose the difference is not significant. (iv) The surprise pattern replicates in-family at 14B. The verdict on inference-time fertility: latent injection buys a content-specific gain in judged novelty that stays below the written word’s, at a coherence cost that halves within 150 tokens and does not vanish. We release the code, the run data with every judge verdict, and the dated pre-registration entries.

1 Introduction

Can a language model produce something genuinely new, and which interventions actually help? A sequence of studies has approached this question one substrate at a time, under a fixed measurement discipline: judged surprise, connection and coherence on windows containing only generated text (so that the judge can never be shown the intervention itself) [21], and, in verifiable domains, an exact evaluator with classical baselines [22]. At the *token* level, anti-probable sampling bought only surface novelty. At the *context* level, periodically interrupting a habituated stream with a new subject raised judged surprise by 1.2 to 1.4 points over habituation alone, on fresh generated-only windows, the flagship positive of the program, while constant pressure (a continuity connective) was worse than nothing [21]. At the *weight* level, consolidating a model’s own verified findings moved the mean toward the known good and never beyond it,

paying for mass with tails [22]; in verified search, operator packages composed additively and stalled at plateaus every arm shared, the flagship one read in the companion’s current version as the resolution limit of the verifier’s grid rather than a construction-family boundary [23]. Between the context and the token sits the object none of these touched: the residual stream $h_\ell(t)$, the thought in progress. Recent work shows, under stated assumptions and empirically on three models, that steered activations reach states no discrete prompt reaches [19], but leaves their *value* untested. Activation steering has mostly been a control tool (removing behaviors, enforcing styles [1, 14, 24, 33, 36]), and where it has been pointed at open-ended generation or at creativity [5, 20, 27] it was not compared with a prompt-matched rival under a judge that never sees the intervention (a *judge-blind* protocol, which is what we mean by deception-resistant below).

This paper does that. Its contributions:

1. **An instrument** (§2): guarded, pulsed translations along model-derived directions in the residual stream (§2), a dimensionless dose unit and the logged effective displacement of every burst; dual MLX/PyTorch backends, bit-identical on the operators. Other operators of the same family (a plane rotor, a projection, repulsion from a moving mean) were tried and are reported as secondaries and screens, not as results.
2. **The temporal contrast** (§3): continuous injection has no fertile band on the plain carrier; on one habituated carrier with the same directions, thirty premises, pulsing beats pressing at equal per-token dose, where the pressed arm collapses coherence, and not at equal integrated dose; the verdict rule and the two-stage extension of that battery are declared with it.
3. **A meaningful direction carries the effect** (§4): sham pulses in random or shuffled directions, which displace the state more, produce less than the real pulse, and a blind manipulation check finds the injected subject in the generated text.
4. **What the pulse buys against the written word** (§4): with both arms judged at the same three offsets after the intervention, the guarded pulse at dose 2.0 trails text interruption in judged surprise (-0.73 on thirty premises, equivalence within ± 0.75 not established), in connection and in coherence, and trails the reset variant of interruption; on the pre-registered cell average, which contains the pulse’s best offset, the two are equivalent within ± 0.75 , with the coherence guard of that primary failing. The coherence cost halves within about 150 tokens and does not vanish. The surprise pattern replicates in-family at 14B (§5).
5. **A spectral description** (§6): across the whole battery, with the rhythm of sentences as the null, the guarded pulse carries the flattest state trajectory; the single-cell “silent orbit” story is reduced to what survives that null. A verified battery in which the same operators changed no verified quantity is reported in §7 as a limitation: it had no headroom.

Every confirmatory analysis below was pre-registered (grid, primary contrast, guards, verdict tiers) and written as code before the data existed; exploratory screens are labeled as such. All deltas are paired per premise, with exact sign-flip tests (2^{10} for ten premises; Monte-Carlo sign-flip for thirty). Intervals: percentile bootstrap 95% CIs over premises for the ten-premise batteries (where the bootstrap under-covers and matches a 90% t interval, so headline ten-premise contrasts also carry t intervals, and inference rests on the exact p values) and for the cell-average contrasts of the thirty-premise equivalence battery (R4b), and t intervals for the other thirty-premise contrasts (the temporal battery R5 and the re-judgings at matched offsets). The decision rules differ by battery and are stated where each result is reported: one primary at $\alpha = 0.05$ for the original batteries; a family of five contrasts with Benjamini–Hochberg q

values for the extension battery R4a; and, for the thirty-premise temporal battery R5, two registered contrasts of which either could confirm the thesis (§3). A coherence guard (median paired coherence change > -1.0) accompanies every primary and is reported with it. The ten confirmatory premises of Ono Filho [21] are reused across this paper’s batteries (they are not held out from anything here); the thirty-premise batteries add the ten original premises of that study and ten expository premises from the same source, three pools whose per-pool means are reported in the released appendix.

2 Instrument and protocol

Operators. All operators are rank-1/2 updates from stored unit vectors, $O(d)$ per position; no $d \times d$ matrix is ever materialized. With $\hat{u}, \hat{w}$ orthonormal and $c_u = \langle h, \hat{u} \rangle$: *translation* $h + \alpha v$; *rotor* $h + \sin \theta (c_u \hat{w} - c_w \hat{u}) + (\cos \theta - 1)(c_u \hat{u} + c_w \hat{w})$, the closed (Rodrigues) form of $\exp(\theta(\hat{w}\hat{u}^\top - \hat{u}\hat{w}^\top))$, verified against the matrix exponential to 10^{-10} ; *projection/dilation* of a subspace; *repulsion* $h + \beta (h - \mu) / \|h - \mu\|$ from an exponential moving average μ of the trajectory. Two composable *guards*: an anchor that restores the coordinate along the premise direction after the operator, and a norm shell that rescales $\|h\|$ to its pre-operator value.

Directions and dose. Concept directions are the model’s own: the unit mean hidden state of a concept text at the target layer (position 0 excluded), differenced against the premise state and normalized, $\hat{v} = (\bar{h}_{\text{concept}} - \bar{h}_{\text{premise}}) / \|\cdot\|$. Doses are expressed in units of the premise-state norm at that layer, $s_\ell = \|\bar{h}_{\text{premise}, \ell}\|$: a translation at dose α adds $\alpha s_\ell \hat{v}$ to every position it touches, so α is dimensionless. §5 carries α and the proportional layer (50% depth) to other models *by convention*, without per-model tuning; whether that convention is optimal there is not tested.

Terms. A *cell* is one 1500-token generation from one *premise* (opening sentence) under one *arm*. The *carrier* is the sampling regime the intervention rides on: *plain* (temperature 1.0, nucleus 0.95) or *habituated* (plain plus the count-weighted log-penalty of Ono Filho [21] over the last 512 tokens, $\ln 1.15$ per occurrence). *Text interruption* appends one of four fixed new-subject sentences to the context every 300 tokens (the injected tokens are excluded from every judged window). The four sentences, rotated in a fixed order, are scene-change openers rather than topics: “That night she dreamed of something else entirely:”, “Meanwhile, in a city with no name,”, “There is an older story about this, and it goes:”, “A question nobody had asked yet:”. A concept direction built from one of them encodes that move of the narrative, not a theme, and “subject” below should be read that way. A *continuous* injection applies the operator at every generated position from token 301 on; a *pulse* applies it for a *burst* of 32 positions at the start of each 300-token *period* (positions $300k+1$ to $300k+32$, $k \geq 1$), with the direction rotating over the same four subject texts used by text interruption; the integrated dose of a pulse is therefore $32/300 \approx 0.107$ of the continuous dose at equal α . *Guards* are the anchor and the norm shell above. The target layer is 18 of the 36 decoder layers of Qwen3-8B-Base (50% depth), chosen on the exploratory map of §3. Because a cell rotates four fixed sentences once each, the verbatim self-copy that dominated the original study’s long runs (65–80% of windows) is largely designed out here (3%).

Carrier and measurement. Unless noted, generation runs on a base model (Qwen3-8B-Base, 8-bit MLX on the origin machine; bf16 PyTorch on GPU for the extension batteries, with the operator algebra verified bit-identical across backends) on the habituated carrier, the regime in which text interruption was established. Judging follows the generated-only protocol of Ono Filho [21]: a judged window is 96 generated tokens; in text-interrupted cells one window

begins 32 tokens after the end of each injected sentence (four per cell) and one 160 tokens after it where the segment allows (capped at three per cell: seven windows), and in every other cell (carrier and state arms, whose interventions leave no token) they sit on a 150-token grid offset by 32 (positions $150g+32$, $g = 1..9$), of which six evenly spread ones are judged, so in state arms two of the six begin exactly at the end of a burst (position $300k+32$) and four begin 150 tokens after one. No window contains injected tokens or overlaps a burst; the judge also sees the 600 preceding tokens as context. The pre-registered primary contrasts use all judged windows; because the two layouts differ, §4 also judges the state arm and the carrier at exactly the text arm’s offsets (+32 and +160 after each burst end) and, after the last of the paper’s five review rounds (reviews by AI reviewer agents, not peer review), the text arm at the end of its injected sentence (+0), which the protocol allows because the sentence sits in the judge’s context and never in the graded window. The judge is Claude Opus 5 (`anthropic.claude-opus-5` on Amazon Bedrock; calls on 30–31 August for the MLX batteries, 1–2 September for the extension batteries and the matched re-judging, 11 September for the +0 re-judging), $k=5$ independent calls per window, per-dimension median, on 0–10 scales. *Surprise* is anchored at 10 = “genuinely startling yet not random”; *connection* scores whether the window joins two distant regions of the preceding text, or an old one with something new, in a way that makes sense (0 when it merely continues one thread); *coherence* whether it reads as one text. The judge’s test–retest behavior was measured in the original study (91% identical medians on byte-identical cells); no cell of this paper was judged twice in duplicate. The judge may refuse degenerate text; refusals are counted per arm and reported wherever they are not negligible. Self-copy (a window reproducing at least half of its 12-token shingles from earlier in the stream) is flagged with the detector of the original study; in the parity battery text interruption copied 3% of its judged windows and the state arms 0%, and estimates on fresh windows only differ from the full estimates by less than 0.1. A cheap coherence proxy (mean NLL of the stream under the same model, untouched) and distinct-4 accompany every cell. Verified experiments (§7) reuse the exact bin-packing evaluator and classical baselines of Ono Filho [22].

3 The temporal contrast

Continuous injection has no fertile band. An exploratory depth×dose map (57 cells, unjudged; the cheap proxies and reading of the text; Fig. 1) located the semantic lever in the middle of the network: early-layer injection collapses generation into memorized register basins (a math-homework attractor), late-layer injection acts as token bias (at high dose the concept’s own token fragment becomes a literal loop), and only mid-network injection transports content. The pre-registered confirmatory then swept the candidate band (layers 14/18/22, $\alpha \in [0.1, 1.5]$, continuous schedule, on the *plain* carrier, 140 cells) with judges. The primary failed in both of its pre-registered parts, the band-core contrast (layer 18, doses 0.25–0.75, the centre of the candidate band) and its coherence guard: band-core surprise minus that carrier’s baseline -0.478 $[-1.478, +0.522]$, one-sided $p = 0.81$, and the coherence guard failed (-1.08). The dose–response falls overall: surprise $2.38 \rightarrow 0.85$ and coherence $4.83 \rightarrow 1.38$ as α rises to 1.5, coherence monotonically and surprise with two small upticks (Fig. 2). More push makes text that a judge finds *less* surprising: degeneration is boring. Notably, the prompt-matched control (the same concept injected as text into this plain carrier) also failed to beat the baseline (surprise 1.87 against 2.38 for the untouched carrier; descriptive). The original study did not find that a written subject needs habituation to work: there, a subject change without habituation scored above habituation alone in judged surprise, and the confirmatory test of whether habituation matters given the interruption was not supported [21]. The control here (one fixed concept sentence, not rotating subject sentences) is not that study’s interruption, and the observation stays descriptive.

The pulse raises judged surprise. The pre-registered transposition of the interruption’s temporal structure (bursts of 32 tokens every 300, directions rotating over the same four new-subject texts used by text interruption, on the *habituated* carrier) raises judged surprise by +0.90 [+0.45, +1.33] over habituation ($p = 0.0059$, coherence guard passed). Text interruption replicates in the same battery (+1.76, $p = 0.0029$), anchoring the instrument. One caution from the extension batteries (§4): on the PyTorch backend the same unguarded pulse gives +0.40 (two-sided $p = 0.13$ at ten premises; +0.41, one-sided $p = 0.014$ at thirty), while the guarded pulse at dose 2.0 gives +1.04 ($p = 0.012$) and +0.93 on thirty premises ($p = 0.0002$). The robust, large form of the finding is the guarded pulse; the unguarded pulse at dose 1.0 is a smaller effect whose size depends on the backend.

The confound, and the one-carrier test. The two results above differ in more than schedule: the continuous battery ran on the plain carrier and the pulse on the habituated one, and habituation by itself changes the stream substantially (it removes the literal loops [21]); the continuous battery also used one fixed concept text for every premise where the pulse rotates the four subject sentences, so the directions differ too. The contrast between them is therefore suggestive, not decisive. The confirmatory that tests it puts both regimes on the habituated carrier with the *same* rotating directions and only the duty cycle varied: pulses of 32 in 300 at $\alpha = 1.0$ against continuous injection at $\alpha = 1.0$ (the per-token dose of the pulse) and at $\alpha = 0.107$ (its integrated dose), paired over the ten premises, one-sided (pulse > continuous). On the ten confirmatory premises the contrasts were a trend (+0.33 [-0.63, +1.30], $p = 0.27$, at equal per-token dose; +0.28 [-0.12, +0.70], $p = 0.13$, at equal integrated dose). The pre-registered extension to thirty premises (same backend, same directions) supports the first: at equal per-token dose the pulse beats continuous injection by +0.48 (95% t interval [+0.01, +0.95], one-sided $p = 0.023$), with the coherence guard passing by a wide margin (+1.70). Two features of that design bound the strength of the verdict and are declared here: its pre-registration admitted confirmation if *either* of the two contrasts reached $p < 0.05$, so the honest threshold for the one that passed is about 0.025 (Bonferroni for two), which 0.023 clears narrowly; and the thirty pairs pool the ten confirmatory pairs already read in the extension battery (a trend, +0.33, $p = 0.27$) with twenty added afterwards. On the twenty premises added after that read, contrast (1) alone gives +0.56 [+0.06, +1.05], one-sided $p = 0.015$, guard +1.78; the result does not depend on the pairs that were seen first. It holds because the continuous arm collapses coherence (-2.42 against the carrier) while leaving surprise where it was (-0.08 [-0.50, +0.35]), and, by the manipulation check, carries the subject into the text (44% of windows) as degenerate prose about it. At equal integrated dose the diluted continuous arm is harmless and not significantly useful (+0.26 [-0.09, +0.60] against the carrier, $p = 0.14$, coherence intact, a positive trend of about the unguarded pulse’s own size), and the pulse’s advantage over it is not significant (+0.15 [-0.19, +0.49], $p = 0.20$). The unguarded pulse itself is a modest, significant effect at thirty premises (+0.41 [+0.04, +0.77], $p = 0.014$). Pressing at the pulse’s dose degenerates; pressing at its integrated dose is indistinguishable from pulsing at this sample size; pulsing raises judged surprise with coherence intact. That is the temporal claim of the paper, and its limit: pulsed beats pressed at equal per-token dose (which is, in the main, a statement that nine times the integrated dose degenerates) and not at equal integrated dose, where the schedule itself would be tested. The plain-carrier dose-response above remains a boundary result.

4 Guards change what a dose does; what the pulse buys against the written word

The unguarded pulse at dose 1.0 pays in coherence (-1.01 [-1.85, -0.23] against text interruption, two-sided $p = 0.049$) while trailing it in surprise (-0.86 [-1.76, +0.02], $p = 0.10$). The pre-

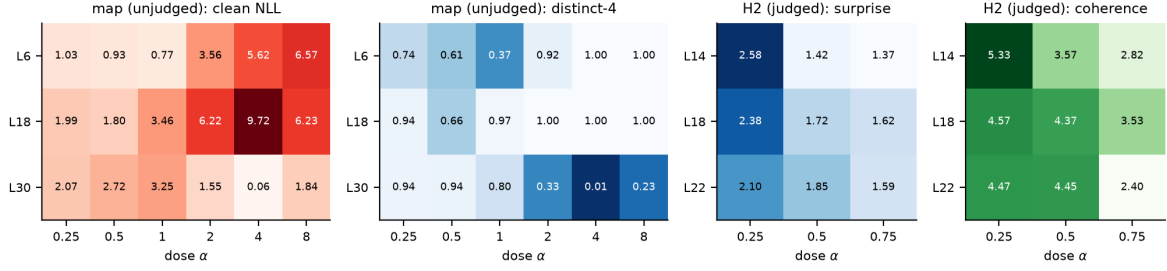


Figure 1: Depth $\times$ dose. Left pair: the exploratory 57-cell map (layers 6/18/30, continuous translation at doses 0.25–8, three premises each, no judge): clean NLL under the untouched model and distinct-4: early-layer injection collapses into low-entropy register basins, late-layer injection at high dose loops the concept’s own token fragment. Right pair: the judged confirmatory grid (layers 14/18/22, doses 0.25–0.75, ten premises each): surprise and coherence, both falling with dose at every depth.

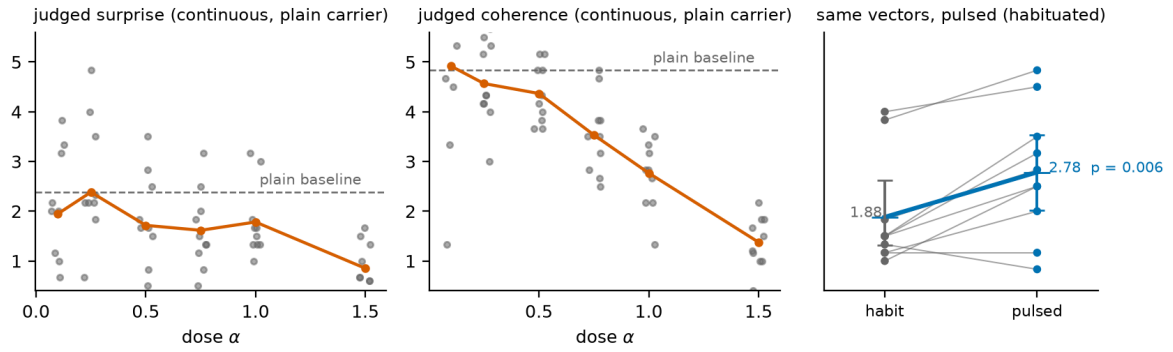


Figure 2: The temporal contrast. Left, center: judged surprise and coherence at layer 18 as continuous dose rises on the plain carrier, against that battery’s baseline (dashed): more push, less surprise, collapsing coherence. Right: the four subject directions delivered as 32-token pulses every 300 on the habituated carrier, against that carrier’s own baseline ($+0.90$, $p = 0.006$). Points are premise means; right, error bars are bootstrap 95% CIs of each mean over premises. The two panels differ in carrier as well as schedule (see the one-carrier test in the text).

registered guard test shows the guards do *not* repair this: at equal dose, anchor + norm shell change coherence by $+0.02$ $[-0.65, +0.63]$ ($p = 0.50$) while weakening the effective push (clean NLL $2.53 \rightarrow 1.83$). Under guards the ladder $\alpha = 1.0 \rightarrow 1.5 \rightarrow 2.0$ climbs monotonically in judged surprise ($2.10 \rightarrow 3.23 \rightarrow 3.40$) at roughly constant coherence (Fig. 4). Because the guards act after the operator, the nominal dose overstates the displacement they leave behind; the extension battery therefore logged the *effective* displacement per burst position, the cosine between the pre- and post-operator state and the relative norm of the change (Table 1). The guarded pulse at dose 2.0 leaves the same mean displacement magnitude as the unguarded pulse at dose 1.0 ($\|\Delta h\|/\|h\|$ 0.75 in both, with different effective directions by construction) and a larger surprise gain ($+1.04$ against $+0.40$ over the carrier; paired over the ten premises, guarded minus unguarded $+0.64$ $[-0.10, +1.38]$, exact two-sided $p = 0.094$, a difference the ten premises do not resolve); the unguarded pulse at 2.0 displaces twice as much (1.53) for less ($+0.86$). So the guards do not “license dose” in the sense of admitting a larger push; they change the composition of the same push (off the premise coordinate, on the norm shell), and that composition is associated with the larger gain.

At the top of that ladder sits the pre-registered comparison against the program’s flagship prompt-side intervention. Guarded pulse at dose 2.0 versus text interruption on the ten premises: surprise 3.40 vs 3.64 ($\Delta = -0.24$, 95% t interval $[-1.11, +0.62]$, two-sided $p = 0.53$), connection 2.38 vs 2.39, coherence within the pre-registered guard. A non-significant difference is not

arm	α	guards	$\cos(h_{\text{pre}}, h_{\text{post}})$	$\ \Delta h\ /\ h\ $	Δ surprise vs carrier
pulse	1.0	none	0.81	0.75	+0.40 ($p = 0.13$)
pulse	1.5	none	0.69	1.13	+0.48 ($p = 0.035$)
pulse	2.0	none	0.59	1.53	+0.86 ($p = 0.043$)
pulse	2.0	anchor + norm shell	0.72	0.75	+1.04 ($p = 0.012$)
sham, random direction	2.0	anchor + norm shell	0.56	0.94	+0.39 ($p = 0.19$)
sham, shuffled direction	2.0	anchor + norm shell	0.56	0.94	-0.09 ($p = 0.81$)
continuous	1.0	none	0.81	0.77	+0.07 ($p = 0.91$)
continuous	0.107	none	1.00	0.08	+0.13 ($p = 0.67$)

Table 1: Effective displacement and its yield. Means over fired positions and ten premises (extension battery, PyTorch bf16); surprise contrasts paired against the habituated carrier, exact two-sided sign-flip.

evidence of equivalence, and the paired standard deviation here (1.21) makes ten premises far too few to show it: the 90% confidence interval, $[-0.94, +0.46]$, does not fit inside any margin narrower than ± 0.95 . The pre-registered extension therefore reran the trio (habituated baseline, guarded pulse, text interruption) on thirty premises on one backend, with an equivalence margin of ± 0.75 fixed in advance (two one-sided tests; ± 0.5 reported as well). The margin is 7.5% of the 0–10 scale, about 60% of text interruption’s own effect over the carrier (+1.22), and about one unit of judge noise per verdict (the original study’s test–retest spread); the reader should note that it admits a deficit of up to 0.7. On the thirty premises the difference is -0.29 $[-0.73, +0.17]$ (two-sided $p = 0.23$), the 90% interval $[-0.68, +0.11]$ lies inside the pre-registered band (TOST $p = 0.028$; power to show ± 0.75 equivalence at the observed paired SD of 1.27: 0.87) and not inside ± 0.5 ($p = 0.18$); removing self-copied windows changes nothing (-0.29 , TOST $p = 0.029$). Both interventions replicate against the carrier on the same thirty premises (pulse +0.93 $[+0.49, +1.36]$, $p = 0.0002$; text +1.22 $[+0.80, +1.65]$, $p < 0.0001$), and the surprise contrast reproduces across backends on the original ten premises (MLX 8-bit -0.24 , PyTorch bf16 -0.29). That primary, however, is favored by the window layout, and three things must be said about it before it is read. First, its coherence guard fails: the median paired coherence change of the pulse against text is -1.02 (mean -0.90 $[-1.20, -0.57]$, $p < 0.0001$), just beyond the -1.0 the paper set for every primary, so by our own rule the cell-average equivalence carries a failed guard. Second, its power of 0.87 is computed at a true difference of zero; at the observed -0.29 it is about 0.61. Third, the judged windows of the two arms do not sit at the same distance from the intervention (§2): two of the six grid windows of a state cell begin at the burst’s end, the pulse’s best offset, and the original text arm had no window there. We therefore judged the arms at the same offsets. Re-judging the state arm and the carrier at +32 and +160 after each burst end (420 windows, registered after the third round of AI review) gives a deficit of -0.34 $[-0.89, +0.20]$ (two-sided $p = 0.21$; TOST ± 0.75 $p = 0.069$, ± 0.5 $p = 0.28$); and judging the text arm at +0, the window that begins at the end of its injected sentence (120 windows, registered after the fifth round of AI review; the sentence sits in the judge’s context, never in the graded window), gives text 5.25 there against the pulse’s 4.03 at the burst’s end ($+1.21$ $[+0.75, +1.67]$, two-sided $p < 0.0001$; connection +0.90, coherence +1.38 in text’s favour). With all three offsets judged in both arms, the fully matched contrast is -0.73 $[-1.19, -0.27]$ (two-sided $p = 0.003$; TOST ± 0.75 $p = 0.46$, ± 0.5 $p = 0.84$; paired SD 1.23), and equivalence is not established. Cell means pool the windows, so the offsets weigh differently in the two arms (four text windows at +0 against the pulse’s two burst-end grid windows); weighting the three offsets equally gives -0.63 $[-1.09, -0.16]$ (two-sided $p = 0.01$; TOST ± 0.75 $p = 0.29$), so equivalence is not established either way. The pulse’s gain over the carrier at the matched +32/+160 offsets is intact ($+0.95$ $[+0.41, +1.49]$, $p = 0.0006$; about +1.0 with the burst-end windows included). The offset curve (Table 2, Fig. 3) shows the shape: relative to the carrier at the same position, the pulse’s gain is +1.20 at the burst’s end, +0.99 at +32 and

arm	judged surprise				connection		coherence			
	burst end	+32	+160	+150	+32	+160	burst end	+32	+160	+150
text interruption	5.25	4.09	3.67	n/a	2.71	2.91	4.85	4.97	4.89	n/a
guarded pulse, $\alpha=2.0$	4.03	3.70	3.40	3.42	2.24	2.23	3.47	3.57	4.02	4.33
habituated carrier	2.83	2.71	2.50	2.62	2.04	1.94	4.87	4.62	4.84	4.97

Table 2: Decay by offset from the intervention. Mean judged scores over thirty premises at windows beginning at the stated offset after the end of the injected sentence (text) or of the burst (pulse; the carrier at the same positions). The +32/+160 state windows are the matched re-judging (after AI review round 3) and the text arm’s “burst end” column is the +0 re-judging (after AI review round 5); the state arms’ “burst end” and +150 (the grid window that begins 182 tokens after the burst’s start, i.e. 150 after its end) come from the original grid. Connection at +0 for text: 3.24 (pulse 2.34).

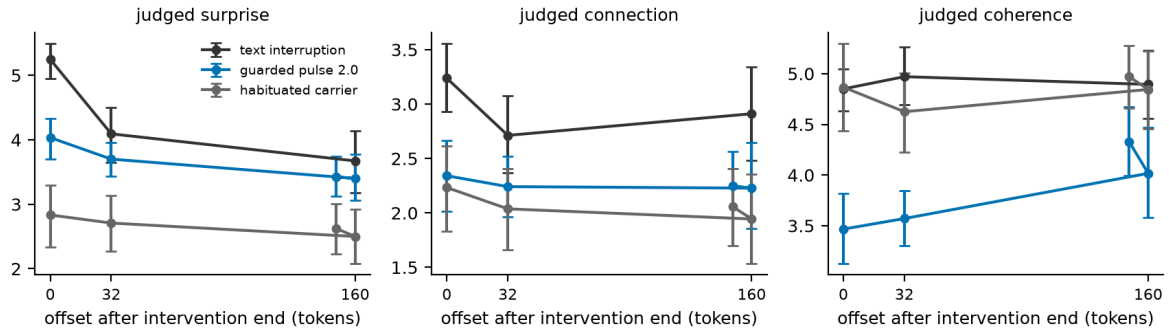


Figure 3: Decay by offset. The three judged dimensions against the offset after the intervention end, thirty premises, bootstrap 95% CIs of the mean over premises. Surprise: the pulse sits below the text at every matched offset, including the burst’s end once the text arm is judged there (+0). The pulse and the carrier carry one further grid point, 150 tokens after the burst’s end. Coherence: the pulse’s cost is front-loaded and halves within the period.

+0.90 at +160; text’s is +1.38 at +32 and +1.17 at +160; the two raw curves fall by similar amounts between +32 and +160 (two points per arm, no uncertainty attached to the slope), and the pulse sits below at every offset. What is short-lived is the cost, not the gain: the pulse’s coherence deficit against the carrier is -1.40 at the burst’s end, -1.05 at +32, -0.82 at +160 and -0.64 at +150 after the end (the last grid window), while text’s coherence never falls below the carrier’s: the cost halves within a period and does not vanish. A per-premise regression of the surprise deficit on the coherence deficit (pulse minus text, 30 pairs) has slope -0.52 ($p = 0.047$): premises where the pulse kept more coherence lost more surprise, and at equal coherence the fitted deficit is -0.75 , descriptive, but a sign that some of the pulse’s surprise is bought with coherence. The other two dimensions do not tie: the pulse trails text interruption in connection (-0.52 [$-0.96, -0.09$], $p = 0.03$) and in coherence (-0.90 [$-1.20, -0.57$], $p < 0.0001$); the “identical connection” of the ten-premise battery did not survive the extension. The claim, then, is one sentence: a subject placed into the model’s stream as pure activation (harder to read off the text than a written one, spending no tokens, continuously dosable) raises judged surprise over the carrier by +1.0 where the written subject raises it by +1.7 (averaged over the same three offsets) and sits below the text at every one of them, at a measurable cost in how connected and how coherent the resulting text reads. What it is not: parity, superiority, or evidence that the state reaches value no prompt reaches (§10).

Does the content of the pulse matter? A pulse could work by carrying a subject or merely by perturbing a habituated orbit. Two pre-registered shams keep everything about the guarded pulse (dose 2.0, schedule, guards, rotation) and replace the direction: a random unit vector per subject, and the subject’s own direction with its coordinates shuffled (same norm

arm	windows	subject named correctly	of windows with a pick	“none”
habituated carrier (base rate)	40	no subject	0/6	85%
text interruption	39	49% [34, 64]	19/25	36%
guarded pulse, $\alpha=2.0$	39	23% [13, 38]	9/14	64%
guarded pulse, $\alpha=1.5$	40	25% [14, 40]	10/21	48%
guarded pulse, $\alpha=1.0$	40	5% [1, 17]	2/5	88%
unguarded pulse, $\alpha=1.0$	39	8% [3, 20]	3/7	82%
repulsion (anchored / not)	40 / 40	2% / 5%	1/4, 2/8	90% / 80%
rotor	39	3% [0, 13]	1/4	90%

Table 3: Manipulation check (parity battery, MLX). A blind judge ($k=1$) names the subject active at each burst-end window; “subject named correctly” is over all windows (Wilson 95% CIs), “of windows with a pick” gives correct/picks, whose null is 25%.

and entry distribution, no meaning in the model’s basis). Both shams displace the state *more* than the real pulse (Table 1) and yield less: the real pulse beats the random-direction sham by +0.65 [+0.13, +1.18] (one-sided $p = 0.029$, BH $q = 0.07$, short of the family threshold) and the shuffled-direction sham by +1.13 [+0.65, +1.65] ($p = 0.001$, $q = 0.005$); against the carrier the shams themselves move +0.39 ($p = 0.19$) and -0.09 ($p = 0.81$). The shams are not less coherent than the real pulse (coherence +0.23 [-0.38, +0.84] and +0.36 [-0.28, +1.01] relative to it), so the judge is not marking them down for damage. The clean result is the shuffled sham, which keeps the direction’s norm and coordinate statistics and loses only its meaning; the random-direction sham is consistent with it. A meaningful direction, not the perturbation of an orbit, carries the effect. Both shams lie outside the set of directions the model itself uses, so they separate meaning from noise; whether *this* subject’s direction does more than another semantic direction would (a fifth subject, another premise’s direction) is not tested here, and a semantic sham matched in displacement is registered as future work. A blind manipulation check completes the picture: for the four windows per cell that begin at the end of each burst or injection (positions $300k+32$; the surprise judge graded two of these per state cell), a separate judge call ($k=1$) names which of the four subjects, if any, the window engages, with the options shuffled per window; text interruption is the positive control and the habituated carrier gives the base rate (Table 3). Two nulls are reported: among the windows where the judge names a subject, chance is 25%; over all windows, a conservative null treats every window as a guess at 25%. The subject reaches the judged text in about half of the text-interruption windows (49% of windows; 19 of 25 picks correct, binomial $p = 10^{-7}$ among picks, $p = 0.001$ over windows) and in about a quarter of the guarded-pulse windows at dose 2.0 (23%; 9 of 14 picks, $p = 0.002$ among picks, not significant over windows), against zero correct picks on the carrier (6 spurious picks in 40 windows); the weak state arms (unguarded pulse, guarded pulse at dose 1.0, repulsion, rotor) transfer nothing detectable (2–8%). Content transfer by the pulse is therefore detectable (significant among picks on the MLX parity battery), dose-dependent under guards, and about half as frequent as by text. In the extension battery, on the other backend, the rates are text 34%, guarded pulse 14% (17 of 53 picks correct, $p = 0.15$ among picks: not significant on this backend), reset interruption 18%, continuous injection at $\alpha = 1.0$ 44% (the subject arrives, in degenerate prose) and the two shams 3–5%, the carrier’s base rate; the full counts and both nulls are in the manipulation-check appendix. The pulse’s judged surprise sits below text interruption’s while it places its subject in the text less often; the shams show that what it does place there is what matters.

The comparator. Text interruption in the preserved context is the strongest prompt-side intervention of the original study *among those that keep the context*; its reset variant (the context replaced by the premise and the new subject at every period) scored as well or better in coherence there. The pre-registered extension includes that reset comparator on the ten

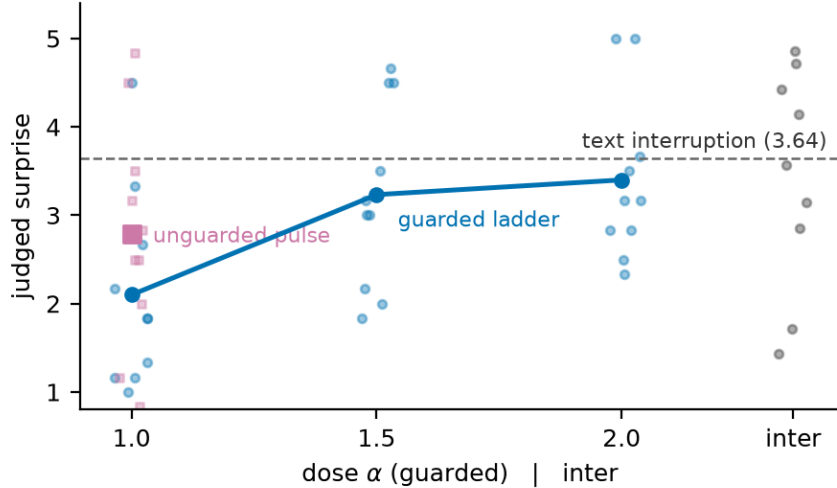


Figure 4: Guards change what a dose does. Judged surprise of the guarded pulse at doses 1.0/1.5/2.0 (points), with the unguarded pulse (square) and the text-interruption reference (dashed line), ten premises each. The ladder reaches the reference within the resolution of ten premises; the equivalence test is in the text.

premises, and it is the strongest arm of the battery: +1.62 over the carrier in surprise with the highest coherence (5.13). The guarded pulse trails it by -0.58 $[-0.99, -0.09]$ (two-sided $p = 0.043$, BH $q = 0.07$) with the coherence guard failing (-1.01). Against the strongest prompt-side intervention we know, then, the state channel is not at parity: it trails the preserved-context interruption at matched offsets (equivalent within ± 0.75 only on the cell average) and trails the reset one.

Two declared secondaries returned honest negatives. A first-version rotor (rotating the premise–subject plane, $\theta = 0.8$) underperforms text interruption (-1.53 $[-2.43, -0.63]$, $p = 0.014$, BH $q = 0.055$; a later cheap screen located its useful range at $\theta \in [1.2, 1.6]$, an untuned operator, not a dead one). And the program’s own registered prediction failed: pulsed state repulsion *without* an anchor did not degenerate (anchored minus unanchored coherence -0.53 $[-1.18, +0.02]$, the wrong sign for the prediction), refusing the fourth appearance of the anchor law (the program’s finding, in weights and in behavior, that repulsion without a fixed reference degenerates); a stress screen then showed why: repulsion from a *moving* mean is self-limiting, because the reference chases the state. The anchor law, as established in weights and behavior, is about fixed references.

5 Generality: the pattern travels in-family

The pre-registered generality battery reran the decisive trio (habituated baseline, guarded pulse at dose 2.0, text interruption) on three untouched models, with the layer at 50% of each network’s depth and the dose carried over unchanged in premise-norm units (a convention, not a calibration).

Qwen3-14B-Base (Hugging Face id Qwen/Qwen3-14B-Base; same family, larger): the primary replicates. Pulse over habituation +1.083 $[+0.567, +1.617]$, $p = 0.0020$, coherence guard clean; against text interruption, a tie of the means at $n = 10$ (on the cell-average window layout, which §4 shows favours the pulse; the 14B cells were not re-judged at matched offsets): surprise 3.48 vs 3.48 ($p = 0.99$; indistinguishable at this sample size, not equivalent). The dimensionless dose and proportional-depth layer worked without retuning on this model.

Qwen3.8-27B (Hugging Face id Qwen/Qwen3.8-27B; a later generation than the

model	pulse – habit (primary)	pulse – inter	verdict
Qwen3-8B-Base (origin, 10 premises, MLX)	+0.90 [+0.45, +1.33], $p = 0.006$	−0.24 [−1.11, +0.62] ^t , $p = 0.53$	indistinguishable
Qwen3-8B-Base (30 premises, PyTorch)	+0.93 [+0.49, +1.36], $p = 0.0002$	−0.29 [−0.73, +0.17], $p = 0.23$ (cell average); −0.73 [−1.19, −0.27] ^t , $p = 0.003$ (matched)	below text
Qwen3-14B-Base	+1.08 [+0.57, +1.62], $p = 0.002$	+0.00 [−0.44, +0.43], $p = 0.99$	tie of means
Qwen3.8-27B (aligned)	+0.20 [+0.02, +0.38], $p = 0.041$	+0.14 [−0.03, +0.31], $p = 0.16$	no reading (carrier breaks)
OLMo-2-13B	−0.09 [−0.32, +0.11], $p = 0.78$	−0.09 [−0.37, +0.11], $p = 0.68$	no reading (register basin; refusals not uniform; text −0.00 too)

Table 4: The generality ladder. Judged-surprise contrasts, paired over the ten premises (thirty in the second row); doses in premise-norm units and layers at 50% depth, carried over by convention without retuning. “Tie” means indistinguishable at $n = 10$, not equivalence. Intervals are percentile bootstraps except those marked t (95% t intervals; for the first row’s, the percentile interval, [−0.94, +0.47], matches a 90% t interval).

Qwen3-8B origin model, aligned): the carrier breaks, not the phenomenon. Under our base-model protocol, which bans EOS and forces 1500 tokens, this chat-trained model opens with fluent prose and collapses into token salad past its natural turn end (judge notes: “degenerate token soup”). A nominally significant pulse effect inside that salad ($p = 0.041$) does not count as support, and we do not count it. Aligned carriers need their own protocol (chat-formatted seeding, free EOS, natural lengths); this is future work, not a footnote.

OLMo-2-13B (different family): no reading. Pulse minus habituation −0.09 ($p = 0.78$). All three arms, however, sit in an encyclopedic register basin (judged coherence 0.4–0.6 everywhere), the OLMo analogue of the register basins of §3, and the judge’s refusals on this model are not uniform across arms: of the scheduled windows, 12% (habituated), 14% (interrupted) and 28% (pulsed) were dropped because every call refused or returned unparsable verdicts, and the surviving windows average 4.4, 4.0 and 4.7 calls rather than 5. Refusals are selective for degenerate text, so the surviving subset is selected differently per arm, most heavily in the pulsed arm; the OLMo contrast is therefore reported without a reading, like the 27B’s. On Qwen3.8-27B the drops were 25/23/22% and on the origin model negligible (6 refused calls in the parity battery). A fourth rung registered in the same battery, gpt-oss-120b, was attempted and abandoned before any cell was judged (mxfp4 kernel failures on the rented GPU). Whether an OLMo null would reflect family-dependence of the phenomenon or miscalibration of layer and dose for this family cannot be separated here; per-model calibration and a degeneration-tolerant judge prompt are registered future work. The generality claim we make is therefore exactly this: *the pattern replicates across scale within a family; family-generality is open*.

6 Spectra: orbits, sentence rhythm, and what flattening survives

Why does the pulse work where pressure fails? A measurement study replayed finished cells through the model, capturing the mid-layer trajectory, and compared the state autocorrelation $A_{\text{state}}(\tau)$ (unit-normed, mean-centered states) against the token autocorrelation (the ground truth of literal loops), reading the strongest detrended local peak of each ($\tau \geq 5$; literal period-2 loops sit below the reader’s floor and are noted as a blind spot).

The prominence ladder in the illustrative cells is clean (Fig. 5). A literal phrase loop shows a state period matching the phrase length ($P = 7$, prominence 0.44). Register collapses (the quiz mode, the math-exercise mode) show matched token–state periodicity at their block rhythms ($P = 123$ and 84). And a habituated cell that repeats no phrase still shows matched token–state periodicity at $P \approx 30$ (prominence 0.18/0.15), whose text reads as sentence-level cycling through the same cast, the “return to the theme, not the sentence” of Ono Filho [21].

Those are single cells, and the population reading is more modest. Replaying every cell of the parity battery (90 cells, 9 arms, 10 premises each) and adding the null a state period has to beat (the autocorrelation of the sentence-boundary indicator, i.e. the rhythm of sentences) gives Table 5. Sentence rhythm is the dominant periodicity in every arm (punctuation prominence

arm	state prominence [CI]	Δ vs habit (p)	punct. prominence	$P_{\text{state}=P_{\text{punct}}}$
habit	0.039 [0.016, 0.070]	n/a	0.132	4/10
inter	0.016 [0.009, 0.026]	-0.023 (0.22)	0.118	3/10
pulse ($\alpha=1$)	0.035 [0.010, 0.080]	-0.004 (0.77)	0.137	4/10
pulseG ($\alpha=1$)	0.054 [0.016, 0.113]	$+0.014$ (0.57)	0.157	1/10
pulseG ($\alpha=1.5$)	0.013 [0.008, 0.018]	-0.027 (0.02)	0.100	3/10
pulseG ($\alpha=2$)	0.010 [0.007, 0.015]	-0.029 (0.006)	0.095	3/10
repelA	0.022 [0.012, 0.038]	-0.018 (0.12)	0.113	4/10
repelN	0.028 [0.010, 0.050]	-0.012 (0.64)	0.115	1/10
rotor	0.078 [0.013, 0.181]	$+0.039$ (0.91)	0.169	3/10

Table 5: Spectra of the whole battery. Dominant detrended local peak of the state autocorrelation per cell; bootstrap CIs over the ten premises; paired differences against the habituated carrier with exact sign-flip p ; the sentence-boundary autocorrelation as the rhythm null; and how often the state period equals the punctuation period within 20%.

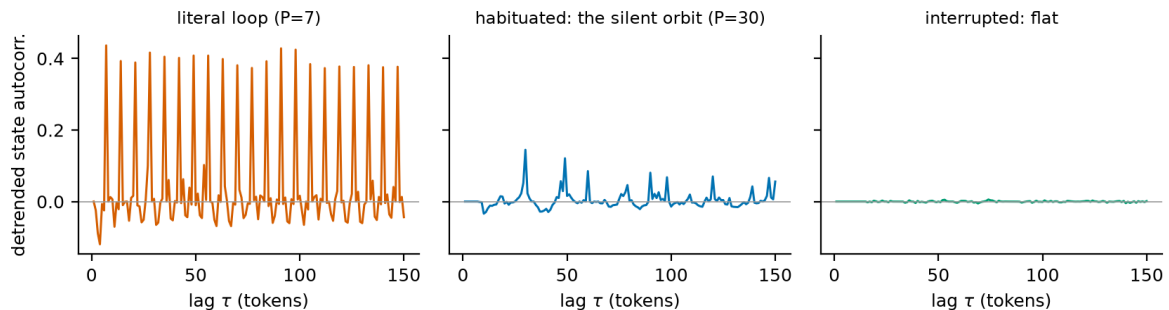


Figure 5: Silent orbits and their removal: three illustrative single cells. Detrended state autocorrelation for a literal loop (state period = phrase length), a habituated cell (the silent orbit at $P \approx 30$, matched in tokens and matched by the punctuation period, see the text), and an interrupted cell (flat). The population contrast for interruption is not significant (Table 5); the arm that flattens at the population level is the guarded pulse.

0.10–0.17); the state prominence of the habituated carrier is small (0.039 [0.016, 0.070]) and its period coincides with the punctuation period in 4 of 10 cells, including the $P=30$ cell above, whose punctuation period is also 30. What survives as an orbit-flattening effect is specific: the guarded pulse at dose 2.0 lowers state prominence against the paired carrier by -0.029 [-0.057 , -0.008] (exact two-sided $p = 0.006$), and at dose 1.5 by -0.027 ($p = 0.02$); text interruption trends the same way (-0.023 , $p = 0.22$); the unguarded pulse, the rotor and moving-mean repulsion do not flatten anything. One more caution: the punctuation prominence also falls in the winning arm (0.095 against 0.132), and per-premise changes in state and punctuation prominence correlate ($r \approx 0.6$), so part of the state flattening is the sentence rhythm flattening. Masking the punctuation period ($\pm 20\%$) leaves a residual state flattening for the guarded pulse at 2.0 (mean -0.023 ; exact sign-flip $p = 0.010$, i.e. nearly every premise moves the same way, while the 95% t interval [-0.048 , $+0.002$] touches zero because two premises carry most of the magnitude) and at 1.5 ($p = 0.02$), not for interruption ($p = 0.08$); the per-arm paired contrasts, the masked residuals and the state–punctuation correlations are tabulated in the released spectral appendix. This section is descriptive, and the exact test is the statement we make. We therefore retract the general form of the claim: the interventions a judge rewards are not all spectral flatteners, and the habituated “silent orbit” is largely the rhythm of sentences. The specific form (the winning state arm carries the flattest trajectory in the battery, part of which survives the sentence-rhythm null) is descriptive, not causal.

arm	candidates	valid	distinct	mean gap	best gap	finds
carrier (none)	293	165	165	+0.0215	−0.00334	2
wildcard pulse	278	121	120	+0.0194	−0.00334	1
domain pulse	308	161	157	+0.0242	−0.00334	3

Table 6: Verified battery, full ledger. Gap = train excess over min(best-fit, first-fit); “find” = a program beating both classics on its variant. Variants with headroom for the carrier: 0 of 16.

7 The verified null, and its floor

Do these operators buy anything a verifier can price? The pre-registered battery ran three arms, untouched carrier, pulses toward narrative subjects (the cross-register wildcard) and pulses toward the domain’s own thinking prompts, with the same configuration as the parity battery (Qwen3-8B-Base on MLX, layer 18, guarded pulse at dose 2.0 with anchor and norm shell, bursts of 32 every 300 tokens, the four narrative directions of §2) inside verified bin-packing notebook generation (16 distribution variants including the far family of Ono Filho [22], 6 notebooks each, 288 notebooks), scoring with the exact evaluator against best-fit and first-fit.

The pre-registered primary is null: per-variant best distance to the classics, wildcard minus control +0.00014 [+0.00000, +0.00036], one-sided $p = 1.0$; transfer and far-family contrasts null; the single best gap across all three arms identical to five decimals (−0.00334). The full ledger explains the identity (Table 6): the untouched carrier had already matched the classics in **16 of 16** variants (reproducing best-fit gives a gap of exactly zero), so the pre-registered measure was saturated by a trivial strategy in every cell; programs that beat the classics do occur (two, one and three finds per arm), but at six notebooks per variant the battery has no power on find rates. The battery therefore does not support a dissociation between judged experience and verified value; it shows that the verified task, at this scale, offers no headroom for any intervention to be measured on. What the ledger does show is a cost: the wildcard arm produced fewer valid programs (121 against 165 for the carrier; paired over the sixteen variants, −2.75 valid programs per variant [−4.69, −0.75], exact two-sided $p = 0.023$, lower in 11 of 16) at the same best value. (An earlier 120-notebook screen had produced a lone wildcard find; the confirmatory dissolved it into sampling noise: all arms find occasionally at $n=6$.) A first-version route-blocking operator (projecting out the top principal directions of a previous solution’s trajectory) behaved as a mild handicap: production down, no depth gained, and a calibration curve showing that the top trajectory components carry the code-writing machinery itself (blocking ≥ 4 directions removes the faculty, not the idea).

Consistent with the attractor results of Ono Filho [22], the verified value reachable by this carrier behaves as a constant that state stimulation did not move; whether it *could* move it on a task with headroom is untested here.

8 Boundaries

For designers of state interventions, the mapped edges: (i) *depth matters more than dose*: early layers destabilize into register basins, late layers are token bias (at high dose the injected concept’s token fragment loops literally); (ii) *guards change the composition of a push, not its size*; (iii) *moving-reference repulsion is self-limiting*: collapse requires a fixed reference, which reframes the anchor law; (iv) *aligned models are a different carrier*: forced continuation past the turn end manufactures degeneration that no operator caused; (v) judged-surprise instruments punish degeneration (they cannot be gamed by noise), but they refuse some degenerate text outright, which costs coverage and must be reported.

9 Related work

Activation steering: what is injected, when, and how it is judged. Adding a vector to the residual stream during generation has a decade of ancestry. PPLM perturbed activations toward a topic during open-ended generation and measured whether the topic reached the text and at what cost in fluency [5], the direct ancestor of our manipulation check; latent steering vectors were extracted from a pretrained model and added to its hidden states to reproduce target sentences [29]; and in-context vectors replace textual demonstrations with a latent shift applied during generation [17], the closest precedent to our question of state against written word, though without a prompt-matched rival. The recent additive family (ActAdd [33], contrastive activation addition [24], truthfulness intervention [14], representation engineering [36], SAE features [4, 31], function vectors [32] and refusal-direction ablation [1]) steers style, safety and factual behavior, and alignment training itself has been read as learning to steer around such directions [11]. Refinements of the additive vector anticipate elements of our instrument: mean-centring corrects the direction for the dataset’s offset [10]; Householder pseudo-rotation edits direction while preserving magnitude [25], the norm-preserving intent our norm shell shares; conceptors replace the additive vector by soft projections onto an ellipsoid, with Boolean combinations [26], which is where our projection operator should be placed; and conditional activation steering [12] gates the edit on a condition, a *when* chosen by the input, where our pulse chooses it by the clock (we know no precedent for clock-scheduled steering, and claim only that we found none). Tan et al. [30] document how unreliable steering vectors are across prompts, which is one reason we report per-premise pairs and effective displacement rather than nominal dose. On creativity specifically, a steering direction extracted from “boring” versus “creative” prompting raises rated creativity when added at inference [20]; Stiefel activation steering explores multiple orthogonal directions for diverse generation [35]; and CreativityNeuro reports that weight-space steering generalises to open creative tasks where activation steering does not [27], a negative for the latent channel that is consistent with, and sharper than, the deficit we measure against the written word. None of these compares the latent channel with a prompt-matched textual rival under a judge that never sees the intervention, and that comparison is what this paper adds. Mishra et al. [19] show, under stated assumptions and empirically, that steered states are unreachable by prompts but do not test their value; our result is one value measurement in that space, and it constrains easy versions of the fertility hypothesis. Weight-space analogues (task arithmetic [9] and the weight steering above) edit the persistent model where we perturb the transient thought.

Repetition and orbits. The loop we measure as a silent orbit has a literal ancestor in the repetition problem: its self-reinforcing dynamics [34], its roots in training data [13], and dedicated neurons [7]. Habituation suppresses the literal form; the state-space rhythm that survives it is what §6 characterizes. Rotations of the residual stream are not exotic to the architecture (rotary position embeddings [28] rotate query-key pairs by position), but ours act on content directions, not positions.

Latent computation. Recurrent-depth and looped transformers move computation from tokens into the residual stream: a learned router decides how many times a token passes through shared blocks [2], or a recurrent block is unrolled at test time to scale reasoning without emitting text [6]. In such models the question this paper asks of the residual stream, what an intervention there buys relative to the written word, stops being a question about steering and becomes one about the model’s primary channel of thought. We note, as speculation rather than a result, that the property we measure (a state injection is harder to read off the text than a written one, §4) would then bear on monitoring; we measured the transfer of a subject to the text, not the detectability of the intervention.

Mechanistic intervention. Causal tracing and model editing record and modify internal routes to *explain* behavior [18]; our route-blocking screen inverts the direction, recording a route to *forbid* it, and reports the first calibration of how much of a trajectory can be blocked before the faculty, not the idea, is removed.

Decoding-time control. Degeneration under maximization [8] anticipates our salad regimes; contrastive decoding [15] and proxy tuning [16] intervene through logits of a second model, a surface channel our late-layer results effectively reproduce (late-layer injection $\approx$ token bias); PPLM [5] is the activation-level counterpart of that family; the mid-network channel studied here is what logit-level methods cannot reach.

The program. Protocol, habituation and interruption from Ono Filho [21]; consolidation, tails and the attractor from Ono Filho [22]; the operator-package factorial from the companion verified-search paper [23]. The automated-researcher study of Chen et al. [3] independently reports the program’s telemetry (method monocultures, flat idea diversity under climbing scores, capability anchors) at industrial scale in an alignment-mitigation setting, a mass regime, complementary to the frontier questions here.

10 Discussion and limitations

What the state channel buys. A new channel into the model’s ongoing thought exists, is dosable and reversible, spends no context, and is harder to read off the text than a written subject. What it buys, measured against the written subject it was modeled on: a gain in judged surprise that sits below the text’s at every matched offset (-0.73 on thirty premises with all three offsets judged in both arms; equivalent within ± 0.75 only on the cell average that contains the pulse’s best offset, and there with a failed coherence guard); carried by a meaningful direction (the shams fail; the manipulation check finds the subject in the text); and paid for in connection and in a coherence cost that is front-loaded, halves within about 150 tokens and does not vanish, with a clear deficit against the reset interruption. This is the state chapter of a four-substrate series (token, context, weights, state), and its verdict on the inference-time fertility hypothesis is the one the abstract states: latent injection buys a content-specific gain in judged novelty that stays below the written word’s, at a coherence cost, not more than the written word. The strong hypothesis, that prompt-unreachable states yield value prompts cannot, remains open; the continuous-injection null and the deficit of the pulse are its first constraining data points. The subject reached the judged text in 14–23% of burst-end windows when injected as state, against 34–49% when written; we note in §9 why that might matter for monitoring, as speculation.

Experience versus value. We had hoped to report a dissociation: state operators changing what a judge experiences while leaving what a verifier prices untouched. The verified battery cannot carry that claim: its measure had no headroom. What survives is weaker and still worth stating: on the one verified task we ran, the operators that a judge cannot tell from the written word produced no verified gain and fewer valid programs.

Limitations. The ten confirmatory premises are reused across this paper’s batteries (each analysis pre-registered before its data, but adaptation risk accumulates and is declared; the equivalence extension adds twenty premises unused by any earlier battery of this paper). The judge is a single instrument (one model family; its test–retest behavior was measured in 21); generality is established across scale, not family; doses and layers were calibrated on the origin

model and carried to others by convention (50% depth, premise-norm units) rather than per-model tuning; the verified domain is one task family and, as reported, had no headroom. The two backends do not give the same carrier: on the ten confirmatory premises the habituated 8-bit MLX carrier scores 1.88 in surprise against 2.48 for the bf16 PyTorch carrier, while the interventions land within 0.1–0.2 of each other across backends (2.78/2.87 unguarded pulse, 3.40/3.52 guarded, 3.64/3.80 text); contrasts against the carrier are therefore larger under quantization, and the unguarded pulse at dose 1.0 is significant on one backend only. The judge was the same model on both, called on different dates (§2), so judge drift and quantization are confounded in that comparison. Bit-identical operators do not imply comparable generation across weight precisions. The plain-carrier dose–response and the habituated pulse differ in carrier, in schedule and in direction; the one-carrier test of §3 is the only clean temporal contrast, and its thirty-premise verdict was reached under a rule that admitted either of two contrasts and by extending ten pairs already read with twenty new ones; the contrast survives on the twenty new pairs alone. The equivalence result holds for judged surprise only, at the ± 0.75 margin only, against the preserved-context interruption only, on the cell-averaged windows only, and with its coherence guard failed; at matched offsets it is not established, and the matched re-judgings were registered after the third and fifth rounds of review by AI reviewer agents, before judging, with the primary result known. All text is English narrative prose from opening sentences; the thirty premises are the ten confirmatory and ten original premises of Ono Filho [21] plus ten expository openings from the same source, and the pulse’s deficit against text is largest in the expository pool (-0.82 in surprise; coherence 3.82 against 4.87), so genre is a variable the design does not control. The layer (18 of 36) and the doses were chosen on the exploratory map of the origin model and carried to the others by convention; dependence on that choice is not mapped. Every cell is 1500 tokens; the batteries ran on one laptop (MLX) and single rented GPUs (PyTorch), with judge costs of the order of tens of dollars per battery: office scale, not a sweep. The mechanism (why 32-token pulses raise judged surprise where constant pressure degrades) is characterized spectrally, not causally.

Future work. Registered and not run: a duty-cycle sweep at equal integrated dose (the schedule test proper); a semantic sham (a fifth subject, or another premise’s direction) and a sham matched in effective displacement; the OLMo trio on a reset carrier and a degeneration-tolerant judge prompt; a re-judging of a sample of MLX cells on the PyTorch judge dates. Beyond that: a protocol for aligned carriers; per-model calibration for cross-family tests; the strong fertility test on a carrier where the pulse and the word are both at ceiling; and substrate-level questions (consolidating the search’s schematic memory rather than its winners, and cross-model state transfer) that the present results motivate but do not touch.

Reproducibility and provenance

The operator algebra, both backends and the generation, judging and analysis scripts are in the code repository <https://github.com/RobertoOno/interrupting-the-loop> as of 10 September 2026; the complete code, including the scripts and appendix sections added after that date, is in an evidence deposit on Zenodo, doi:10.5281/zenodo.23163299. The cells and judge verdicts of every battery, including the matched-offset re-judging of §4, are in the repository’s release **run-data-v2**, archived as Zenodo 10.5281/zenodo.22714138; the +0 re-judging, made after that release, the run logs and the dated pre-registration entries, excerpted from the project notebook (in Portuguese), are in the evidence deposit. All confirmatory analyses were committed as code before their data existed. Table 7 lists every battery behind a number in this paper.

battery (section)	carrier / backend	cells	judged windows	status	runs / script	appendix
depth×dose map (§3)	plain / MLX 8-bit	57	0 (proxies only)	exploratory	state_band, state_screen / state_inject.py	none
continuous band, H2 (§3)	plain / MLX 8-bit	140	846	pre-registered	state_h2 / run_h2.py	H2
parity battery: pulse, guards, H1 trio (§3-4)	habituated / MLX 8-bit	90	550	pre-registered	state_pulse / run_pulse.py, run_pulse2.py	PULSE, PULSE2, H1
manipulation check on the parity battery (§4)	same cells, one judge call	90	356 ($k=1$)	pre-registered	state_pulse / manip_check.py	MANIP
rotor and repulsion screens (§4)	habituated / MLX 8-bit	21	0 (proxies only)	exploratory	state_pulse / state_inject.py	none
verified screen B-1 (§7)	code carrier / MLX	4 runs	verifier	exploratory	state_codeSmoke / state_code.py	none
verified B-CONF (§7)	code carrier / MLX	288 notebooks	verifier	pre-registered	state_bconf / run_bconf.sh, analysis_bconf.py	BCONF
route-blocking screen (§7)	code carrier / MLX	3 runs	verifier	exploratory	state_rb / state_code.py	none
spectra (§6)	replay of the 90 parity cells	90	n/a	measurement	spectral / spectral.py, spectral_summary.py	SPECTRA
generality, Qwen3-14B-Base (§5)	habituated / PyTorch bf16	30	190	pre-registered	pod/gen_qwen14 / torch_state.py, analysis_gen_ladder.py	GEN5
generality, Qwen3.8-27B (§5)	habituated / PyTorch bf16	30	146	pre-registered	pod/gen_qwen38 / same	GEN5
generality, OLMo-2-13B (§5)	habituated / PyTorch bf16	30	156	pre-registered (no reading)	pod/gen_olmo / same	GEN5
generality, gpt-oss-120b (§5)	n/a	0	0	pre-registered, abandoned (kernels)	none	none
extension R4a: slams, one-carrier, reset, dose (§3-4)	habituated / PyTorch bf16	80	487	pre-registered	r4 / run_r4_pod.sh, analysis_r4.py	R4
extension R4b: equivalence on 30 premises (§4)	habituated / PyTorch bf16	90	569	pre-registered	r4 / same	R4
extension R5: continuous arms on 30 premises (§3)	habituated / PyTorch bf16	60	357	pre-registered (either-of-two rule)	r4 / same	R4
matched-offset re-judging of pulse and carrier (§4)	same cells, re-judged	60	420	registered after AI review round 3, before judging	r4 / r4_matched_judge.sh	R4
text arm judged at +0 (§4)	same cells, re-judged	30	120	registered after AI review round 3, before judging	r4 / r4_text0_judge.sh	R4

Table 7: Provenance. Each battery with its carrier and backend, number of cells, judged windows (up to $k=5$ judge calls each; the surviving-call counts per arm are in the appendices), its status, the run directory under `runs/` and the script, and the appendix (`docs/APPENDIX_*.md`) that reproduces its numbers. Dated pre-registrations are entries of the project notebook (in Portuguese), excerpted in the evidence deposit (doi:[10.5281/zenodo.23163299](https://doi.org/10.5281/zenodo.23163299)), which also holds the +0 re-judging; the other cells and verdicts are in the release `run-data-v2` (Zenodo [10.5281/zenodo.22714138](https://zenodo.org/record/22714138)).

Acknowledgements and tooling

The experiments, analyses and text of this paper were produced by the author working with an AI assistant, Claude (Anthropic), inside the Claude Code environment: the assistant wrote and ran the code, operated the runs under the author’s direction and drafted the manuscript (the Claude Fable 5 and Claude Fable 5.1 models, 27 August to 11 September 2026, for versions 1–3; Claude Opus 5.5 for the corrections of version 4). Every design decision, result and claim was reviewed by the author. The five review rounds mentioned in the text (1–11 September 2026) were reviews by AI reviewer agents, not peer review; the paper has not been peer reviewed. The judge was run on Amazon Bedrock; generators ran locally on Apple silicon via MLX and on rented GPUs via PyTorch.

Changes in version 4

Corrections only; no new data, and no headline number changes. (1) §4, the reset comparator: a sentence left over from version 1 said the pulse is equivalent in surprise to the preserved-context interruption; it now states the paper’s result (it trails at matched offsets and is equivalent within ± 0.75 only on the cell average). Text interruption is called the program’s flagship, not its strongest, prompt-side intervention (abstract, §4), since the reset variant is the strongest arm of the battery. (2) The fully matched contrast: how its windows weigh the offsets is stated, and weighting them equally gives -0.63 (the headline -0.73 stands; equivalence is not established either way); the $+0.95$ gain over the carrier is scoped to the $+32/+160$ offsets. (3) Content transfer by the pulse is significant among picks on the MLX battery only (on PyTorch, 17 of 53 picks, $p = 0.15$). (4) The 14B model is Qwen3-14B-Base, and its tie with text is marked as measured on the cell-average layout. (5) Interval types: the cell-average contrasts of the thirty-premise equivalence battery carry percentile-bootstrap intervals, as computed, not t intervals as §1 stated; Table 4 now gives the premises and the interval type of each row. (6) §3: the claim that the plain-carrier control reproduced a finding of Ono Filho [21] (that novelty injection needs habituation) is withdrawn; that study’s confirmatory test did not support it. (7) The companion verified-search paper is described and cited in its current version (arXiv:2609.29636; the flagship plateau read as a resolution limit of the grid). (8) Availability: the run data and the dated pre-registration entries are in an evidence deposit; the earlier release lacked the +0 re-judging, and an English digest of the notebook that version 3 pointed to does not exist. (9) The use of an AI assistant is declared, and the review rounds are described as reviews by AI reviewer agents.

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